

Agri Price Prediction System

AI-Powered Crop Market Forecasting for Smarter Farmer Decisions

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PROBLEM Problem Statement

Indian farmers lose **20-30% of potential income** due to poor market timing. With over 7,000 APMCs across the country, price volatility driven by seasonal demand, rainfall patterns, and arrival quantities makes it nearly impossible for smallholder farmers to decide *when* and *where* to sell their produce. Existing tools lack hyperlocal, week-by-week price forecasting — leaving farmers dependent on middlemen and guesswork.

SOLUTION Proposed Solution

To solve the issue of income loss caused by poor market timing, we have thought of developing an **ML-powered crop price** forecasting system that helps farmers identify the best time and market to sell their produce. The model studies historical mandi prices, rainfall trends and seasonal demand across different APMCs to predict future prices with better accuracy. The solution would provide local, week-by-week price forecasts, enabling farmers to make informed selling decisions instead of depending on middlemen or assumptions. By using machine learning and data analytics, the system aims to reduce uncertainty in agricultural markets and help smallholder farmers maximize their profits. Our objective is to create a smart, accessible, and data-driven platform that empowers farmers with timely market insights and improves their overall financial stability.

METHOD Methodology

- 1

Data Collection
APMC market data (2020-2024): prices, arrivals, rainfall across multiple states and commodities.
- 2

Feature Engineering
Temporal lag targets created via grouped shifts. Categorical encoding via OneHotEncoder for Commodity and Market.
- 3

Model Training
Four separate Random Forest Regressors (n=100 trees) trained — one per forecast week — on pre-2024 data.
- 4

Prediction & UI
Streamlit dashboard enables commodity/market selection; models return 4-week price outlook with sell recommendation.

ARCH System Architecture

DATA INPUTS

Commodity | Market | Harvest Date

FEATURE ENGINEERING

OneHot Encoding + Temporal Lag Targets (Week 1-4)

ML MODELS (x4)

Random Forest Regressors — one per forecast week

FEATURES Key Features

- 4-Week Forecast:**
Separate model per week for higher accuracy
- Rainfall-Aware:**
Incorporates real rainfall data as a price driver
- Market-Specific:**
Predictions scoped to individual APMC markets
- Sell Recommendation:**
Highlights the best week with % gain estimate
- Simple UI:**
Streamlit app

STAGE Current Stage of Development

- DONE

Four Random Forest models trained & serialised
- DONE

Streamlit prototype deployed locally
- ACTIVE

Integrating harvest-date aware rainfall lookup
- NEXT

PPT Presentation

IMPACT Expected Impact

Empowers smallholder farmers with **data-driven selling decisions**, reducing dependence on intermediaries and improving price discovery across rural India.

15-25%

Income Increase
for Smallholders

7,000+

APMCs
Scalable To

4 Weeks

Forecast
Horizon